

TELECOM CHURN INTELLIGENCE

Customer Churn Prediction

Predictive Strategy Based on the Device Lifecycle for Telecommunications Operators

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Market Context



\$1.55T USD

Projected global telecommunications revenue for 2026, operating at an average EBITDA margin of 35%.



Loyalty Crisis

Approximately 77% of consumers report no loyalty to their current carrier and are willing to switch when presented with a better technical offer.



15%–25% Churn

Average annual churn rate in postpaid plans, making churn a major operational and financial leakage point for the sector.

| The Business Problem

- **Purely Reactive Retention:** Interventions happen when the customer has already decided to leave, increasing reacquisition cost up to 7 times compared with preventive retention.
- **Mass Campaigns Without Focus:** Broad offers cannibalize revenue by giving discounts to customers who, statistically, would not have left anyway.
- **Underused Data Assets:** Mobile device obsolescence is not being leveraged as a relevant early signal of the customer's switching intent.

Hidden Financial Leakage

Blind retention is not sustainable. Early, targeted intervention makes it possible to intercept the customer shopping window before competitors trigger the switching decision, protecting Lifetime Value (LTV).

| Dataset and Methodology

Dataset Dimensions

- Volume: 100,000 unique customer records.
- Features: 100 variables (82 numerical, 18 categorical).
- Balance: The target variable is nearly balanced, providing statistically adequate volume to model both retention and churn.

Methodological Treatment

- Categorical Variables: Strict imputation with the "Unknown" label to avoid biased assumptions.
- Missingness Flags: For binary numerical variables, mode imputation was combined with missing-value flags to preserve the meaning of absence.
- Split: Stratified 80/20 Train/Test partition to preserve the population distribution.

ANALYTICAL HYPOTHESIS

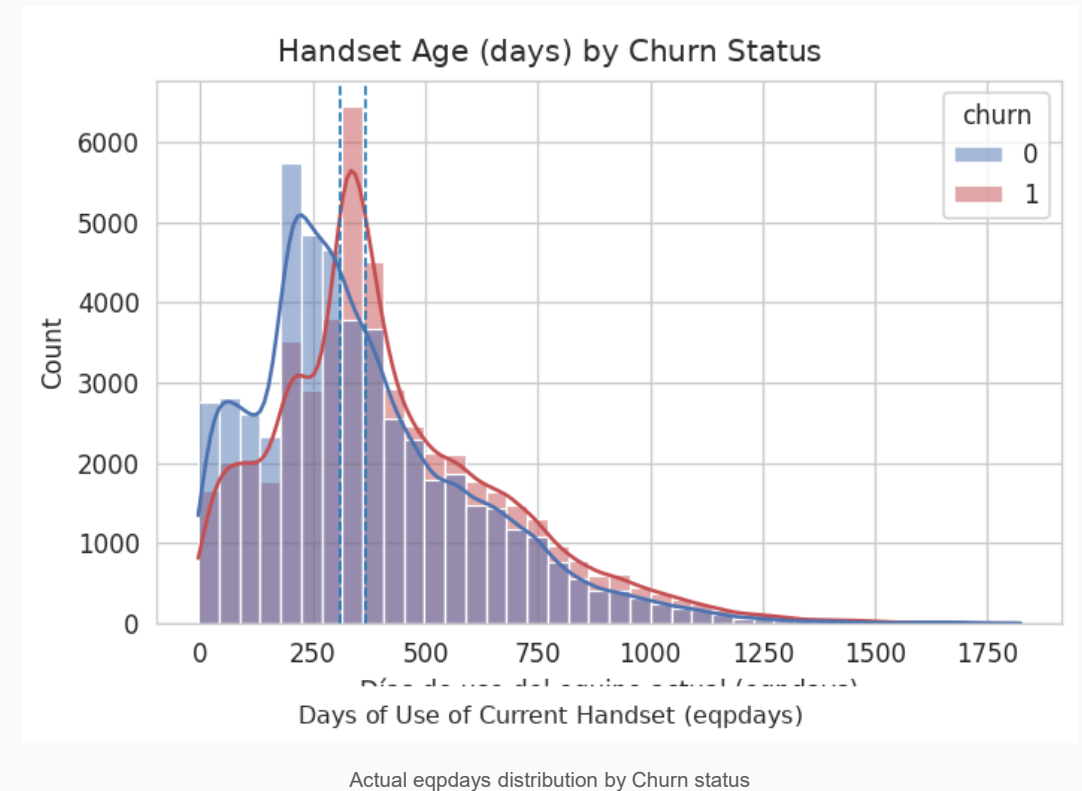
Mobile handset age is one of the key predictors of churn

"As the customer's device ages, the customer may enter a natural technology renewal window. The 700-day threshold will be treated as an initial operational hypothesis, not as a definitive causal rule, and must be validated through the pilot before automating retention offers."

Exploratory Data Analysis (EDA)

Relevant Statistical Findings:

- Handset Age: The churn segment shows a higher average device age, consistent with the renewal cycle.
- Structural Overlap: The distributions of both classes overlap substantially; eqpdays should not be interpreted as a standalone cause.
- Multivariable Approach: Correlation was used as exploration, confirming the need for a robust predictive model.

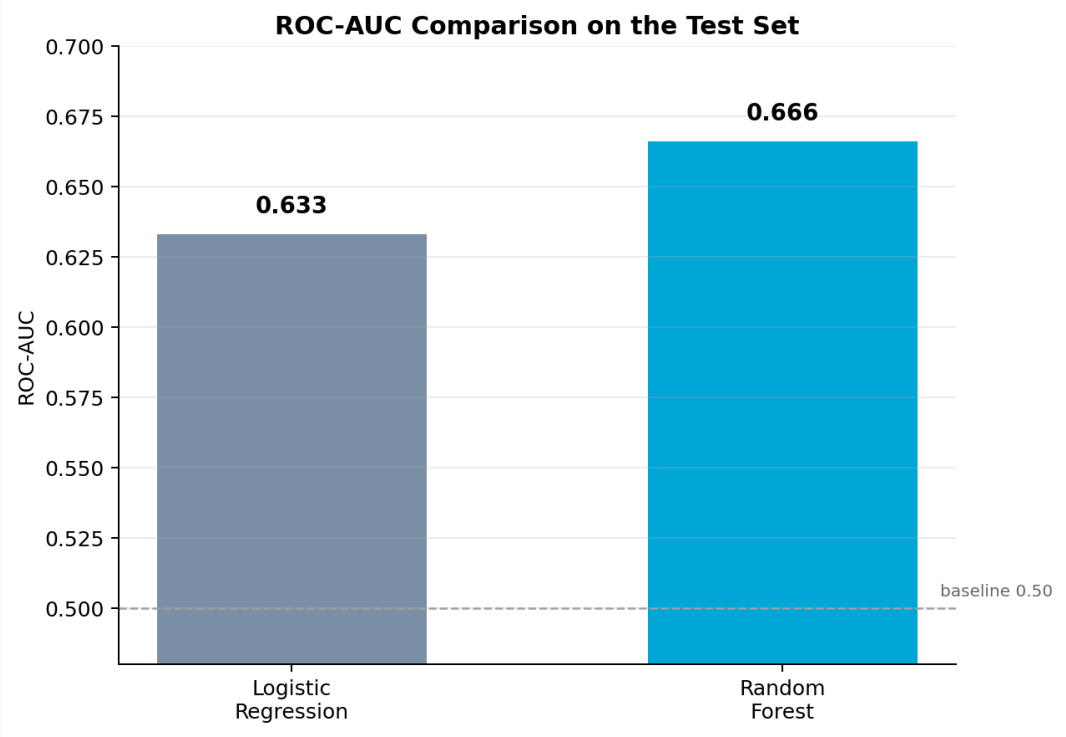


ML Model Evaluation

Comparative evaluation on the test set (20K customers).

Algorithm	Accuracy	Recall	F1 Score	ROC-AUC
Logistic Regression	59.6%	59.4%	59.3%	0.633
Random Forest ✓	61.5%	63.6%	62.1%	0.666 ▲

The Random Forest model achieved stronger Recall and ROC-AUC performance, making it better suited than linear approaches for prioritizing customers with higher churn risk.

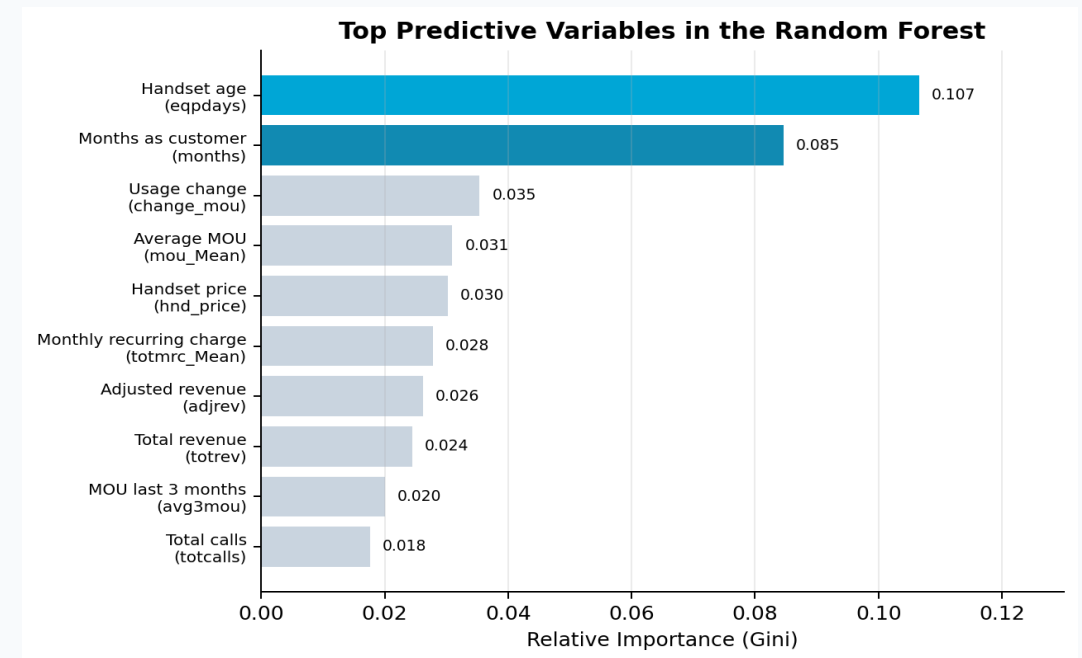


ROC-AUC Comparison: Logistic Regression vs. Random Forest

Feature Importance

Main predictors (Gini Importance):

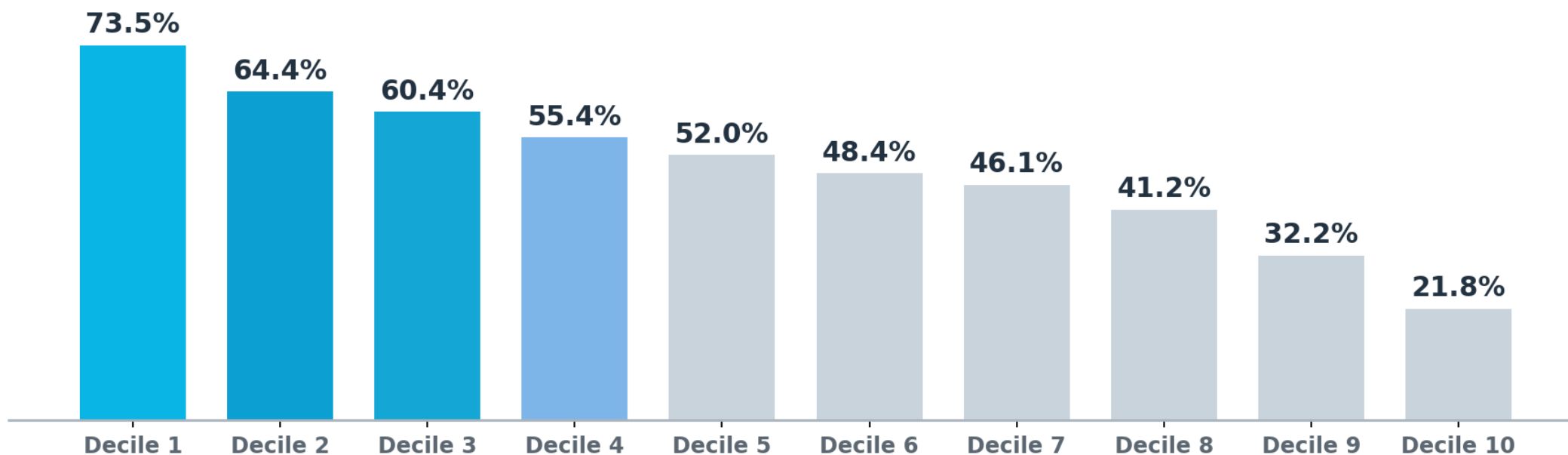
- Handset Age (eqpdays): The variable with the highest relative weight in the Random Forest.
- Service Tenure (months): Complements the device lifecycle to better identify the renewal window.
- Usage and Revenue (change_mou, mou_Mean, adjrev): Add behavioral and economic value signals to the score.



Grouped and readable Random Forest variable importance

Risk Decile Analysis

This stratification shows that the upper-risk groups capture an actual churn rate far above the population average.



The highest-risk Top 20% (Deciles 1 and 2) concentrates actual churn rates well above the average, optimizing the retention focus.

| Assumptions and Financial Impact (ROI)

☰ Business Assumptions

- Active Base: 1,000,000 subscribers.
- ARPU (Average Revenue per User): \$50 USD/month (\$600 USD/year).
- Base Churn Rate: 20% annually (~200,000 customers leaving).
- Campaign Cost: \$70 USD per contacted customer.
- Effectiveness (Save Rate): Conservative 35% on contacted customers.

📊 Impact Comparison

Reactive Campaign (No Model)

Blind application to 200K customers.
Projected cost: $200K \times \$70 = \$14.0M$ USD.
Spend is diluted by giving incentives to users who are not necessarily at real risk.

Predictive Campaign (Prioritized Top 4%)

Operational subsegment of 40K customers with the highest probability within the high-risk pool.
Cost: $40K \times \$70 = \$2.8M$ USD.
Gross protected revenue: $40K \times 35\% \times \$600 = \$8.4M$ /year.
Estimated net impact: \$5.6M/year (2.0x net ROI).

TECHNOLOGY STRATEGY

| DLIS System

We introduce the Device Lifecycle Intelligence System (DLIS): an automated platform that predictively links hardware aging with the customer's latent intention to leave the mobile network.

We transform Random Forest analytics into an operational tool that enables the CRM to execute retention campaigns around the estimated critical window, using handset age as an adjustable and pilot-validatable hypothesis.



| DLIS Operating Flow



1. Daily Ingestion

Consolidation of data usage logs, billing metrics, and continuous updates of the device lifecycle (*eqpdays*).



2. Predictive Score

Automated algorithmic processing (Random Forest) to classify and rank the customer base into 10 risk deciles.



3. Targeted Action

API connection with the CRM to activate focused offers, prioritizing the initial Top 4% and scaling to the Top 20% based on observed ROI.

Implementation and Strategic Value

DLIS Timeline (6 Months)

Phase 1 (Months 1–2): Foundations

Data ingestion, rigorous preprocessing, and initial model training.

Phase 2 (Months 3–4): Controlled Pilot

A/B tests isolating the impact on 10% of the population base.

Phase 3 (Months 5–6): Full Scale-Up

Final integration with Marketing tools and automated retraining.



EXECUTIVE CONCLUSION

Churn is not a surprise event; it is a predictable variable

Through the DLIS System, we turn the inertia of the smartphone technology wear cycle into a valuable tactical advantage for protecting the customer mobile Lifetime Value (LTV).

Formal Next Steps

Approval of the pilot plan (Phase 1) and technical orchestration of corporate Data Warehouse APIs.

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